Matteo Tusoni

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About me

PhD student in Computer Science at Sapienza University of Rome, working at the intersection of machine learning, causal discovery, time series analysis, and agent-based modelling. My research focuses on developing AI methods for understanding complex dynamical systems, with applications to financial and macroeconomic systems.

Education

PhD in Computer Science *Nov 2024–Present*
Sapienza University of Rome Rome, Italy

- **Description:** Research on machine learning, causal discovery, time series analysis, and agent-based modelling for financial and macroeconomic systems.
- **Advisor:** Prof. Novella Bartolini.

MSc in Theoretical Physics *Oct 2020–Mar 2023*
Sapienza University of Rome Rome, Italy

- **Description:** Thesis on Higgs-boson decays into $\bar{b}b$ and $\bar{c}c$ with the ATLAS detector, including the development of machine learning methods for particle-flow reconstruction.
- **Advisor:** Prof. Marumi Kado; **Co-advisor:** PhD Francesco Armando di Bello.
- **Grade:** 110/110 cum laude.

Research Experience

Department of Computer Science, University of Oxford *Feb 2026 – May 2026*
Visiting researcher Oxford, UK

- **Research:** Developed and analyzed differentiable approaches to agent-based model simulations for economic and financial systems.
- **Team:** Research team led by Prof. Michael Wooldridge and Prof. Anisora Calinescu.

Applied Research Team, Bank of Italy *Jan 2025 – Aug 2025*
PHD internship Rome, Italy

- **Research:** Studied causal inference techniques for macroeconomic time series, focusing on structural dependencies in high-dimensional financial systems.
- **Research:** Explored multi-agent models for macroeconomic simulation and emergent behaviour in complex adaptive economies.

CERN – ATLAS Experiment *Mar 2022 – Mar 2023*
Member, internship Geneva, Switzerland

- **Research:** Contributed to AI-based particle detection, detector simulation, and studies of the Higgs boson within the ATLAS collaboration.
- **Contribution:** Contributed to ATLAS Note ANA-HIGG-2020-20-INT1 on Higgs-boson decays into $\bar{b}b$ and $\bar{c}c$ in VH production.

INFN – Frascati National Laboratories *Mar 2021 – Aug 2021*
Member, internship Frascati, Italy

- **Research:** Researched plasma acceleration and electron-beam interaction with ionized gas within the SPARC_LAB collaboration.

Publications

C=Conference/Workshop, J=Journal, U=Under review

[U] **Micro-Level Priors Enable Macro-Level Learning of Neural Agent-Based Models** *2026*
Under review – First author

[C] **Robust Causal Discovery in Real-World Time Series with Power-Laws** *2026*
ICML 2026 – Spotlight. Top 2% of papers – First author

[C] **Causal Discovery for Stochastic Power Law Time Series** *2025*
AIFIN 2025, ECAI Satellite Workshop – First author

[C] **Placy – Power law causal Discovery** *2025*
Complexity in Economics & Finance, StatPhys29 Satellite Workshop – First author

[J] **Reconstructing particles in jets using set transformer and hypergraph prediction networks** *2023*
European Physical Journal C – Co-author

[J] **Configurable calorimeter simulation for AI applications** *2023*
Machine Learning: Science and Technology – Co-author

[C] **HGPflow: Particle reconstruction as hyperedge prediction** *2022*
Machine Learning and the Physical Sciences, NeurIPS 2022 Satellite Workshop – Co-author

Awards, Selected Programs & Grants

Machine Learning Summer School NYC 2026 *Jun 2026*
Columbia University New York, USA

- Fully funded participant in a competitive two-week residential programme in advanced machine learning.

EcoIDEA – Eco-friendly Inference in Distributed Learning *May 2026 – Present*
Project Member Sapienza University of Rome

- **Project:** Sapienza-funded research project on energy- and resource-efficient inference in distributed AI systems.
- **Funding:** €95,363.23 awarded
- **Scientific coordinator:** Prof. Salvatore Pontarelli.

G-Research – Spring into Quant Finance *Apr 2026*
G-Research London, UK

- Fully funded participant, selected among the top 1% of applicants for a programme in quantitative finance and research.

Sapienza's Call for Scientific Research Grant *Jan 2026*
Sapienza University of Rome Rome, Italy

- Competitive seed funding for an independent research project.

Professional Experience

Senior Data Scientist *May 2024 – Nov 2024*
Ernst & Young S.p.A. Rome, Italy

- **Data Science:** Delivered end-to-end data science projects for major clients, including Costa Crociere and Barilla, developing statistical and machine learning models and optimization methods for complex business problems.
- **Leadership:** Coordinated and managed the data science team within the group, contributing to the development of its technical capabilities and expertise.

Data Scientist & AI Specialist *Jun 2023 – May 2024*
Leonardo S.p.A. Rome, Italy

- **Secure & Explainable AI:** Developed a platform for secure and explainable AI, supporting the evaluation and interpretation of machine learning models.
- **Quantum Computing:** Designed and developed a traffic simulation using a quantum annealing algorithm to minimize traffic congestion.
- **GCAP:** Developed AI models for the Global Combat Air Programme (GCAP), a sixth-generation combat aircraft programme involving Italy, the United Kingdom, and Japan.
- **EDOCC:** Contributed to the definition and development of AI capabilities for the European Defence Operational Collaborative Cloud (EDOCC), an EU-funded federated cloud platform for multi-domain military operations.

Technical Skills

Programming	Python, C++, C, R, Julia, MATLAB, SQL
Machine Learning	Deep learning, PyTorch, TensorFlow, scikit-learn, Keras
Methods	Causal discovery, causal inference, time series analysis, agent-based modelling
Tools	Git, Docker, ROOT
Operating Systems	Linux, macOS, Windows
Languages	Italian (native), English (advanced), French (basic)